

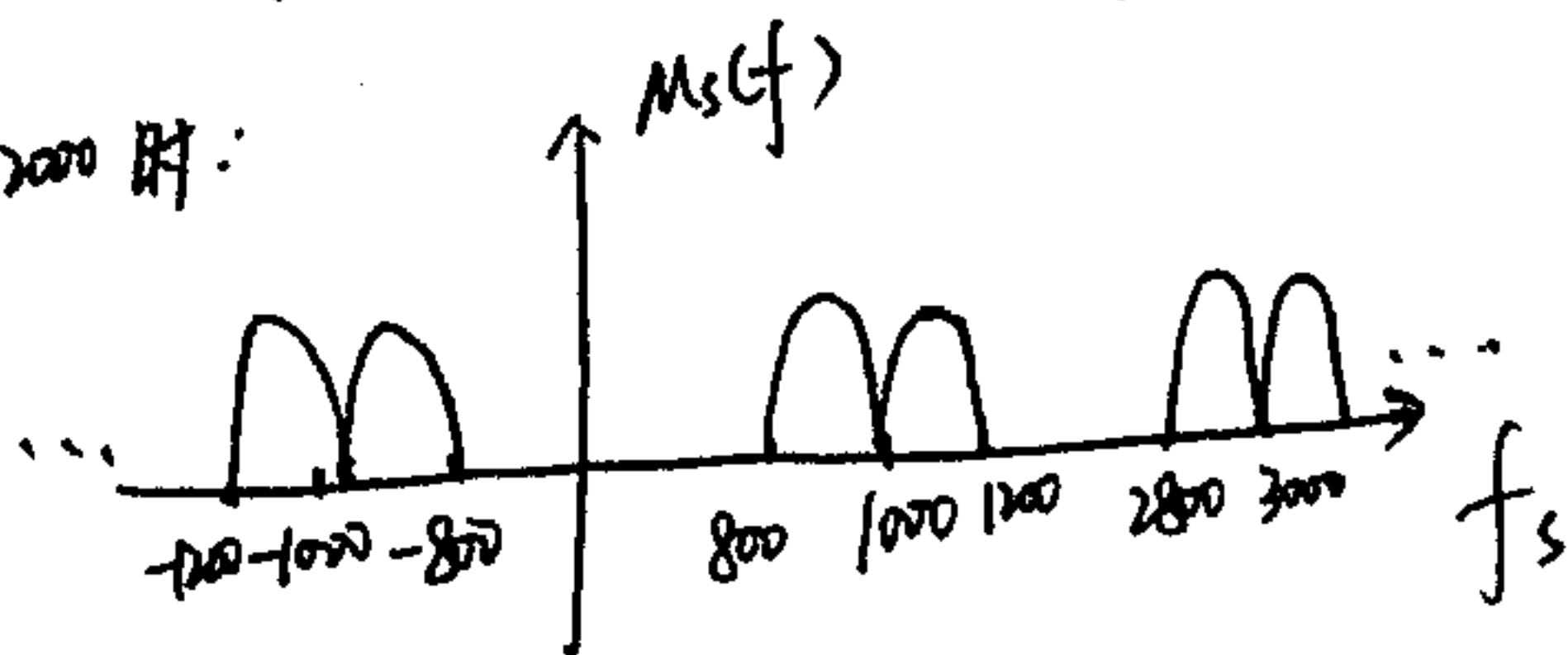
4-1.

(1) $f_s \geq 2f_H$

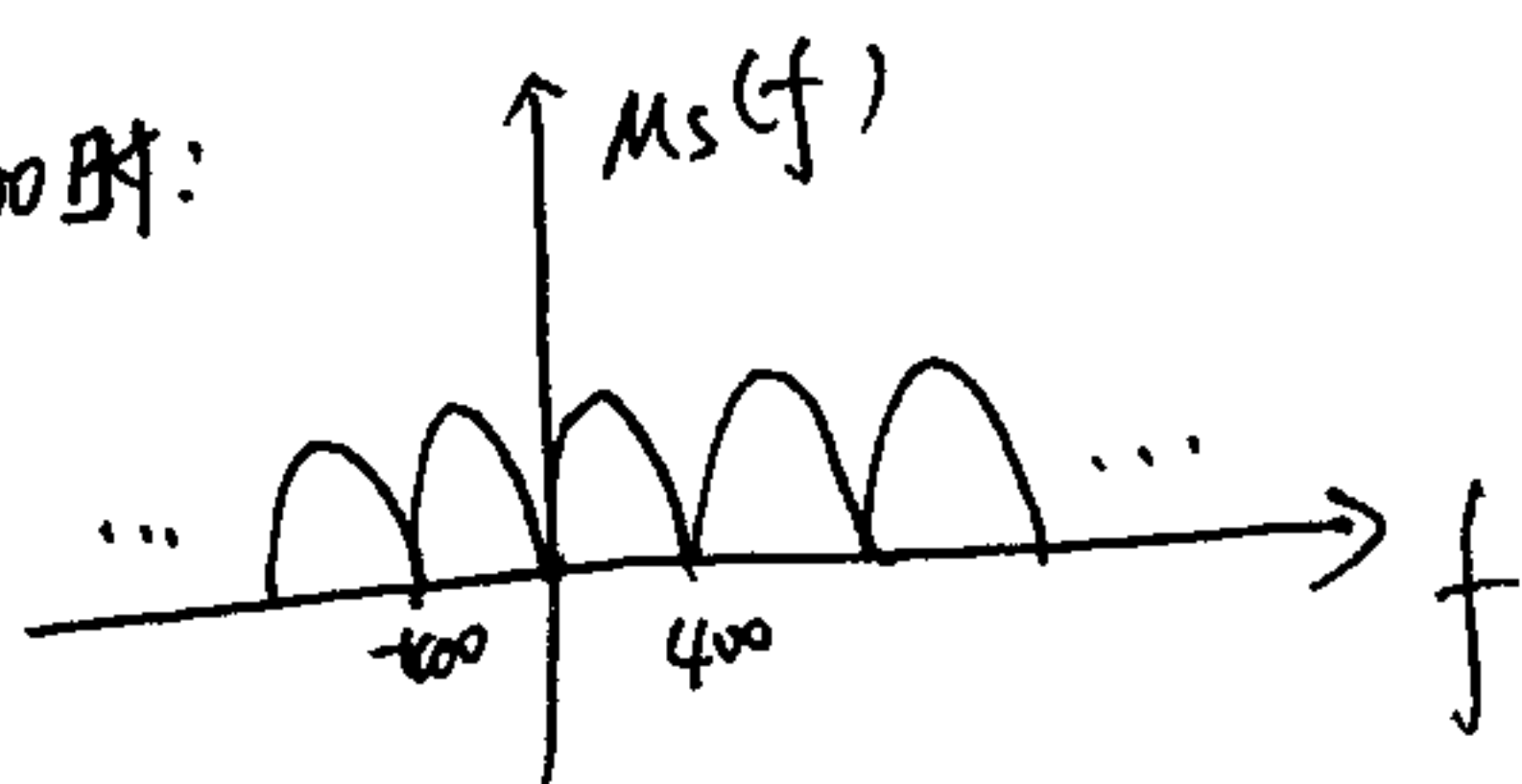
$\therefore f_{s(\min)} = 2000 \text{ Hz}$

(2) 带通信号: $f_{s(\min)} = \frac{2f_H}{K} = \frac{2000}{K}$
 $K = \frac{1000}{200} = 5 \quad \therefore f_{s(\min)} = 400 \text{ Hz}$

(3) 当 f_s 为 2000 时:



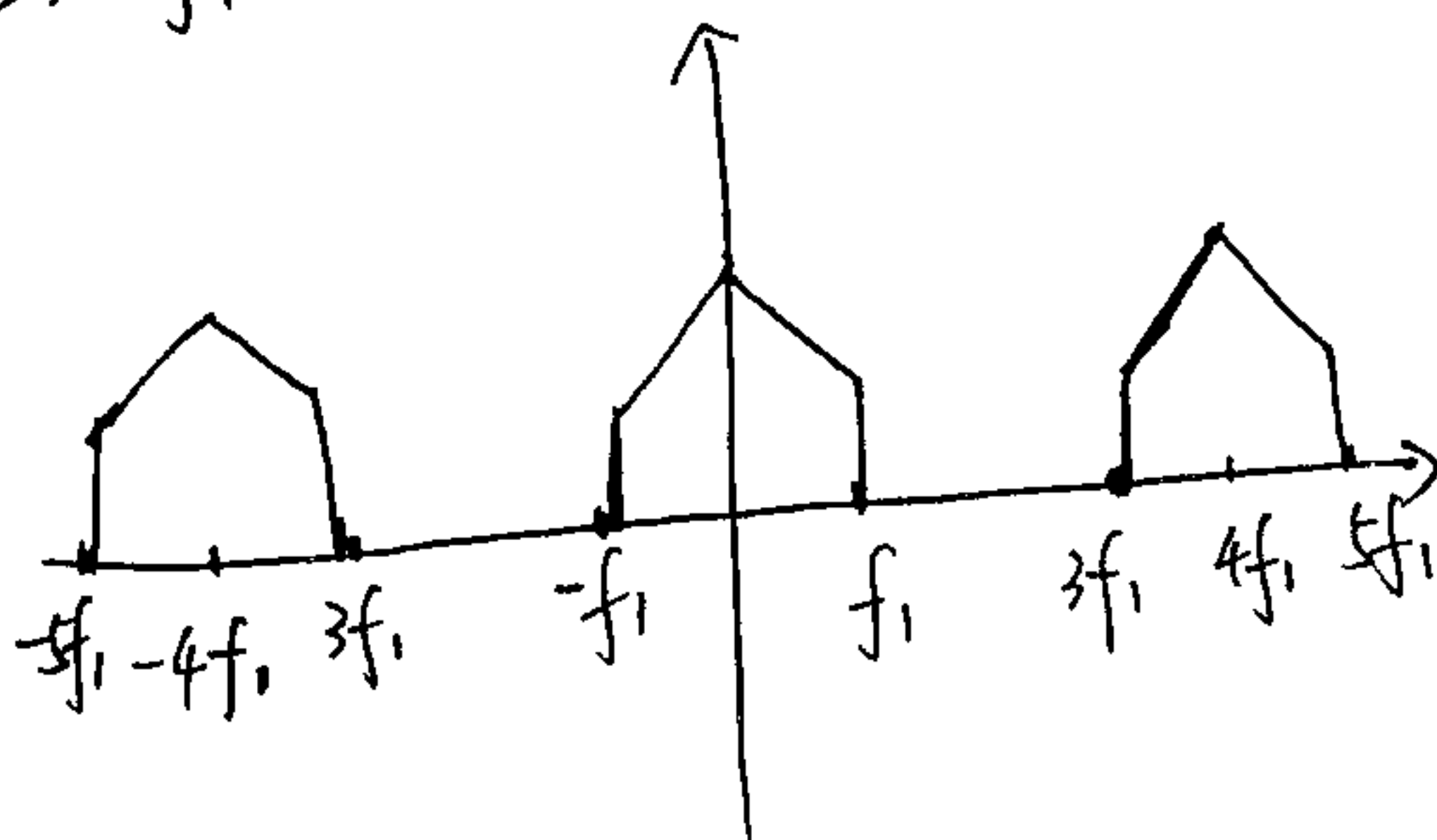
当 f_s 为 400 时:



4-3.

(1) $f_s \geq 2f_1$

(2)



(3). 采用接收滤波器:

$$H_2(f) = \begin{cases} 1/H_1(f) & |f| \leq f_1 \\ 0 & \text{其他} \end{cases}$$

4-10

(1) $\Delta = 2 \text{ mV}$, 抽样值为: 398Δ

极性码 $C_1 = 1$,

段落码: 落在 6 段, 段落码为 101

段内码: $398 = 256 + 8 \times 16 + 14$, $C_5 C_6 C_7 C_8 = 1000$

$\therefore$ 8 位码为: 11011000

(2). 量化电平: $256 + 8 \times 16 + 8 = 392\Delta$

量化误差: $398\Delta - 392\Delta = 6\Delta$

①

$$(3). \text{时} (392)_{10} = 256 + 128 + 8 = 2^8 + 2^7 + 2^3$$

$\therefore$ 12位线性幅度码: 001100010000

4-13.

$$(1) \text{量化级数: } M = \frac{5 - (-5)}{0.01} = 1000$$

量化噪声的平均功率:

$$N_q = \frac{1}{2} \sum_{i=1}^M \int_{m_i}^{m_{i+1}} (x - q_i)^2 p(x) dx = \frac{M \Delta^3}{24a} = \frac{10^{-4}}{12}$$

$$\text{信号功率: } S = \int_{-5}^5 x^2 \frac{1}{10} dx = \frac{10^2}{12}$$

$$\therefore \frac{S}{N_q} = 10^6$$

$$(2). \text{量化电平: } 256 + 14 \times 16 + 8 = 488 \Delta \approx 1.19V$$

当 量化电平 $480\Delta < x < 496\Delta$ 时, 都编码为 1011110

即 $1.17V < \text{电平} < 1.21V$ 时, 都编码为 1011110

$$\therefore \text{发生概率: } p = \frac{1.21 - 1.17}{10} \approx 0.39\%$$

4-18.

$$(1) f_{s(\min)} = 2 \times 7K = 14K$$

$$(2) R_b = n \cdot b \cdot f_s = 10 \times 3 \times 16000 = 480 K (\text{bit/s})$$

4-19.

(1)

TS0	TS1	TS2	TS3	TS4	TS5	TS6	TS7	TS8	TS9	TS10
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↓
帧同步时隙

$$\text{帧长: } 1/8000 = 125 \mu s$$

$$\text{bit数: } 11 \times 8 = 88 \text{ bit}$$

每帧有11个时隙

$$(2) R_b = 8K \times 8 \times 11 = 704 K \text{ bit/s}$$